**SOFTWARE ENGINEERING LAB 1**

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| NAME: | PREMA P KOTUR |
| SRN: | PES1UG24CS343 |
| SECTION: | F |

**Requirements Specification Table**

*Problem Statement #36 — Restaurant Table Booking & Pre-Ordering App*

*Lab 1: Requirements Engineering & UML Use-Case Modelling*

| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| **FR-001** | Functional | The system shall allow diners to pick an open table from an interactive 2D floor plan and link pre-ordered menu items to the reservation. | High | Pass: Table is reserved and a kitchen pre-order ticket is queued. Fail: The same table is double-booked for an overlapping dining window. | Core booking functionality; directly enables the seamless arrival-dining goal stated in the problem context. |
| **FR-002** | Functional | The system shall allow diners to browse the digital menu and pre-order food items to be associated with their reserved time slot. | High | Pass: Selected items appear in an itemized order summary linked to the reservation ID and are timestamped for kitchen prep. Fail: An order is accepted without a valid reservation reference. | Pre-ordering is the second half of the core value proposition; without it the kitchen cannot prep ahead of arrival. |
| **FR-003** | Functional | The system shall calculate and display an estimated kitchen preparation timeline based on the pre-ordered items and current kitchen load. | Medium | Pass: An estimate is shown to the diner within 3 seconds of order submission and is accurate to within +/-5 minutes of the logged actual completion time. Fail: No estimate is shown, or the estimate is off by more than 10 minutes in 3 consecutive test orders. | Sets accurate diner expectations and helps synchronize arrival time with food readiness. |
| **FR-004** | Functional | The system shall allow the Restaurant Manager to view and | High | Pass: A status change made by the manager is reflected on the floor-plan dashboard | Gives staff operational control of the floor and is |

| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| | | update the real-time status of each table (Available, Reserved, Seated, Billed). | | immediately after the update action. Fail: The dashboard shows a stale status after the update. | required to keep the live floor plan (FR-001) accurate. |
| **FR-005** | Functional | The system shall allow diners to cancel a reservation up to 30 minutes before the booked time slot, releasing the table and voiding the associated pre-order. | Medium | Pass: A cancelled reservation frees the table slot for new bookings and the diner receives a cancellation confirmation. Fail: The table remains blocked, or the pre-order ticket is still active in the kitchen queue after cancellation. | Prevents tables from being locked by no-shows and gives diners reasonable flexibility. |
| **NFR-001** | Nonfunctional (Performance & Security) | Table status changes (Reserved, Seated, Billed) must sync across all manager floor terminals in under 500 ms. | High | Pass: Benchmark tests confirm the target latency and security standards are met under simulated peak load. Fail: Sync latency exceeds 500 ms, or the sync channel fails an authentication/encryption check. | Multiple staff members act on the same floor plan simultaneously; stale data causes double-seating errors. |
| **NFR-002** | Nonfunctional (Reliability / Availability) | The system shall be available 99.5% of the time during restaurant operating hours (11:00-23:00 daily). | High | Pass: Uptime monitoring logs show at least 99.5% availability over a rolling 30-day period. Fail: Downtime exceeds 0.5% of operating hours in any calendar month. | Reservation and pre-ordering are business-critical during service hours; downtime directly causes lost bookings and revenue. |

Actors

1. Diner — primary end user who books tables and pre-orders food.
2. Restaurant Manager — staff user who manages table status and floor operations.
3. Payment Gateway (external system) — authorizes and processes reservation payments/deposits.
4. Kitchen Display System (external system) — receives pre-order tickets and prep-timeline data.

Use Cases

1. UC-01  View Live Floor Plan

2. UC-02 Reserve Table (<<include>> UC-03, <<include>> UC-04)
3. UC-03 Pre-Order Food Items (<<include>> UC-06)
4. UC-04 Process Payment (extended by UC-05)
5. UC-05 Apply Discount / Loyalty Offer (<<extend>> UC-04)
6. UC-06 Calculate Kitchen Preparation Timeline
7. UC-07 Manage / Update Table Status (extended by UC-08)
8. UC-08 Cancel Reservation (<<extend>> UC-07)